PROBLEM STATEMENT SIH26167

SatQuery AI

Detailed engineering pipeline for an interactive vision-language assistant over multimodal remote-sensing imagery

Problem statement ID	SIH26167
Title	SatQuery AI — An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
Organisation	Indian Space Research Organisation (ISRO)
Department	Department of Space / ISRO
Category / Theme	Software · Space Technology
Idea submission deadline	30 September 2026
Adaptation dataset	BigEarthNet.txt (Sentinel-1 SAR + Sentinel-2 MSI + text); any open-source training data also permitted
Public benchmarks	VRSBench, RSVQA (single-image) · CDVQA (change VQA)
Final evaluation set	ISRO/SAC — pre-georeferenced, co-registered Cartosat-2S optical + RISAT SAR pairs; annotations undisclosed

Scope of this document

Seven-stage pipeline covering ingest through evaluation, with the data contract at each boundary, the model registry, the auditable execution trace, and the ranked technical risks.

Problem statement text retrieved from the official portal (sih.gov.in/sih2026PS) on 7 September 2026. Engineering choices below are proposals, not requirements: requirements traceability is given in section 2.

Compiled 7 September 2026

1. What the problem statement actually demands

Three clauses in the statement drive the engineering more than the feature list does. Each one is load-bearing for a design decision taken later in this document.

1.1 The evaluation sensors are not the training sensors

Adaptation is mandated on BigEarthNet, which is Sentinel-1 and Sentinel-2 at 10 m ground sample distance. Final scoring uses Cartosat-2S optical, which is sub-metre, paired with RISAT SAR. That is a 15-30x resolution gap, across sensors with different band counts and a different SAR calibration convention. Train at 10 m, get scored at 0.65 m.

Figure 1 — The mandated adaptation corpus and the final evaluation set occupy different ends of the resolution range. Closing this is a Stage 0 and Stage 2 concern, not something to patch at the end.

Consequence

Every scale-sensitive decision is designed around this gap: ground sample distance is a first-class field on the tile contract (Stage 0), a conditioning input to the backbone (Stage 2), and a selection key in the tool registry (Stage 1). A held-out high-resolution non-Sentinel split is the only pre-eval signal available, so it is built before the models are.

1.2 Only the observable trace is scored

The statement is explicit that the controller may plan internally but that “only the observable execution trace, including the selected task, models or tools, permitted parameters, and outputs will be evaluated,” and that “internal reasoning text is neither required nor evaluated.”

A chain-of-thought agent therefore earns nothing scoreable. A deterministic router that emits a complete, well-formed trace scores identically and fails far less often. This is why Stage 1 is a classifier plus a validation gate rather than an LLM planner, and why the trace record in Stage 4 is the single source of truth behind the UI, the JSON view and the PDF report.

1.3 Scores are normalised before combining

Metrics across tasks are normalised and then combined, so a weak task cannot hide behind a strong one and a zero anywhere is expensive. Breadth of working coverage beats depth on any single task. The evaluation harness in Stage 6 is therefore built early and run continuously, so the weakest task is always known.

1.4 Mandatory functional floor

Requirement	Status in statement	Where satisfied
Remote-sensing adaptation of at least one visual or VL component	**Mandatory**	Stage 2 — shared backbone, QLoRA on BigEarthNet
Single-image visual question answering	**Mandatory**	Stage 2 — VQA head
One further single-image task: captioning or grounding	**Mandatory** (either)	Stage 2 — both are built; VRSBench scores both
Change description or change-based VQA from a bi-temporal pair	**Mandatory**	Stage 2 — change + change-QA
Spatial change map	Optional, where reference masks exist	Stage 2 — change mask is produced regardless; it feeds change-QA

Requirement	Status in statement	Where satisfied
Complementary extraction from a co-registered optical-SAR pair	**Mandatory**	Stage 2 — fusion branch
Agentic selection, sequencing and execution of specialists	**Mandatory**	Stage 1 — controller
Input upload and compatibility checking	**Mandatory**	Stage 0 + Stage 1 gate
Visual evidence, confidence, execution summary, downloadable report	**Mandatory**	Stage 3 + Stage 4
Generic LLM or VLM without remote-sensing adaptation	**Explicitly insufficient**	Adaptation is the Stage 2 backbone, not a prompt

2. Pipeline overview

Seven stages. Stages 0 through 4 run per query; Stage 5 hosts them; Stage 6 measures them. Each stage publishes a typed contract, named on the right below, and no stage reaches around another to inspect raw inputs.

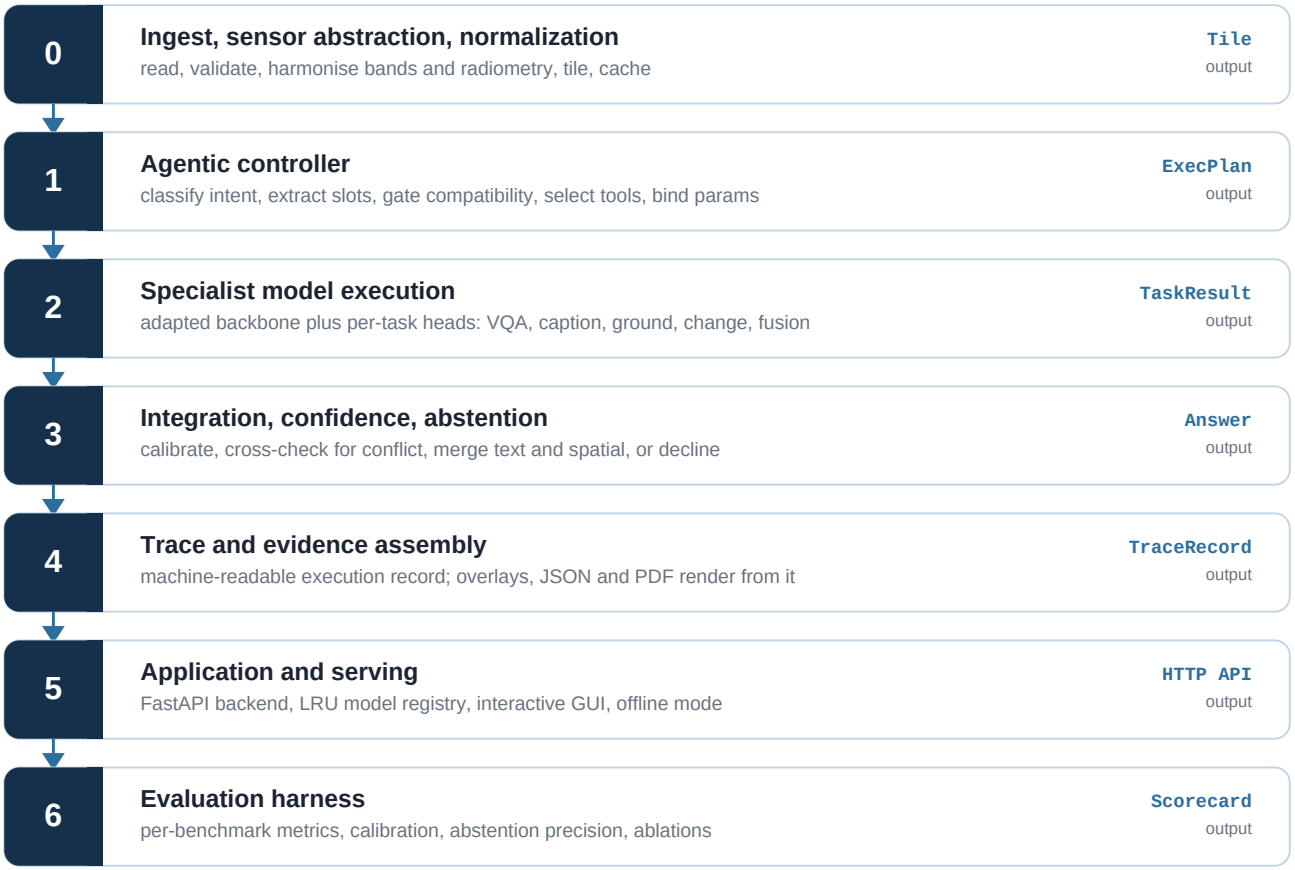

Figure 2 — Stage chain with the contract each stage emits. Contracts are the integration surface: a stub and a trained checkpoint are interchangeable as long as the contract holds.

2.1 Per-query runtime path

The statement enumerates six controller duties. They map one-to-one onto the runtime steps below, and each emits its own trace line, so the observable record mirrors the required list exactly.

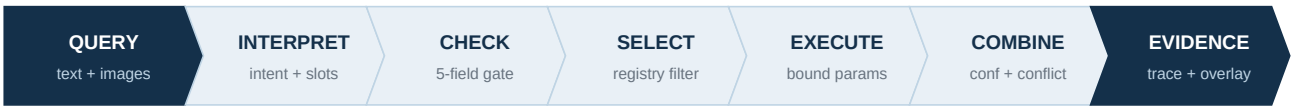

Figure 3 — Runtime path. A failed gate short-circuits to a reasoned abstention, which is still a complete trace and still scoreable.

2.2 Six supported intents

Intent	Representative query from the statement	Inputs	Output kind
vqa	open question over one image (mandatory baseline)	1 tile	text, categorical
describe	“Describe the land-cover and major objects visible in this image.”	1 tile	text
ground	“Highlight the water body referred to in the query.”	1 tile	geo box, mask
change	“What changed between these two dates, and where did the change occur?”	bi-temporal pair	text + change mask
change-qa	“Has the built-up area increased, decreased, or remained unchanged?”	bi-temporal pair	categorical + areas
fuse	“Use the optical and SAR images together to identify built-up and water-covered regions.”	optical + SAR pair	text + class masks

The hard boundary is **change** against **change-qa**. Both are bi-temporal, so lexical routing conflates them, yet one wants a description plus a map and the other wants a three-way categorical backed by areas. Router training data is weighted toward that boundary.

3. Stage 0 — Ingest, sensor abstraction, normalization

This is the only stage that touches raw files, and it is the stage that absorbs sensor heterogeneity so nothing downstream has to. Inputs span band counts from 1 to 12, ground sample distances from 0.15 m to 10 m, and two unrelated SAR sensors.

Source	Modality	Bands	GSD	Role
BigEarthNet	S1 GRD + S2 MSI	2 + 12	10 m	adaptation (mandated)
VRSBench	optical RGB	3	sub-metre	caption, ground, VQA eval
RSVQA LR / HR	Sentinel-2 / USGS	3-13	10 m / 0.15 m	VQA eval
CDVQA (SECOND)	optical RGB pairs	3	sub-metre	change-VQA eval
LEVIR-CD, OSCD	optical pairs	3 / 13	0.5 m / 10 m	change training
Cartosat-2S + RISAT	**optical PAN/MX + SAR**	**1 or 4, + SAR**	**~0.65 m**	**final ISRO/SAC scoring**

Do not branch on dataset anywhere downstream

Cartosat-2S PAN is a single band and MX is four bands. Neither is RGB, neither is Sentinel-2's twelve. Any model that assumes three or twelve channels breaks on evaluation day. Variability belongs in a sensor profile registry, so adding RISAT is one profile entry and a calibration check rather than a code change.

3.1 Sensor profile registry

```
SensorProfile:
  id                # s1_grd | s2_msi | cartosat2s_pan | cartosat2s_mx | risat_sar | generic_rgb
  platform, native_gsd
  band_map          # semantic slot -> physical index, or None if absent
                    #   blue green red nir swir1 swir2 pan  VV VH HH HV
  dtype, scale, offset, nodata
  radiometry        # optical_percentile_stretch | sar_linear_to_db
  speckle_filter    # refined_lee | none
  incidence_angle   # required for SAR normalisation; differs S1 vs RISAT
  calibration_note  # free text, surfaced in the trace
```

Ship profiles for S1, S2, Cartosat-2S PAN, Cartosat-2S MX, RISAT, and a generic-RGB fallback for the PNG and JPEG benchmark path the statement permits.

3.2 Normalizer sequence

Ordered, and each step logs to the trace:

- **Open and validate.** Read with rasterio. Reject non-georeferenced GeoTIFF with a typed error, unless the file is on the whitelisted benchmark path. The statement allows PNG and JPEG only for prescribed benchmarks, so enforcing that as a rule is free compatibility-check credit.
- **Resolve profile and map bands.** Physical indices to semantic slots. Missing slots stay explicitly None. Never zero-fill silently, or a Cartosat panchromatic scene is read as a very dark RGB.
- **SAR path.** Linear power to dB, refined-Lee speckle filter, per-scene normalisation, incidence-angle correction from the profile.
- **Optical path.** Keep the raw float array for models. Compute a separate 2-98 percentile stretch for display only. Display radiometry must never reach a model.
- **Quality mask.** Sentinel-2 SCL where available, otherwise a small cloud classifier. This is what later *justifies* a SAR fallback rather than merely asserting that fusion helps.
- **Tile.** 512 x 512 with overlap, carrying ground sample distance through.
- **Cache.** Canonical tiles to disk, so offline mode is a cache hit and not a second code path.

3.3 Canonical tile contract

```
Tile:
  id, sensor_id
  bands: dict[str, ndarray]   # semantic slots; absent slots simply missing
  gsd                         # first-class, not metadata -- Stage 2 conditions on it
  crs, transform, bounds      # transform is what makes grounding geo-anchored
  ts                          # acquisition timestamp
  qa_mask                     # cloud / shadow / nodata

SceneMeta: parent_scene, tile_grid, source_path, ingest_params, profile_id
```

3.4 Pair handling

Two pair types with different invariants, both checked here and gated in Stage 1:

Pair type	Checks performed	Failure handling
Cross-modal (optical + SAR)	resample to a common grid; footprint IoU; assert the two sensors really are different modalities before allowing fusion	degrade to single-modality with an explicit flag in the trace
Bi-temporal	distinct timestamps; same footprint; phase-correlation alignment with the **residual shift logged in pixels**	abstain if residual exceeds tolerance, naming the measured shift

The ISRO set arrives pre-co-registered, so the residual will be near zero there. Benchmark pairs and user uploads will not be, and at 0.65 m a sub-pixel misalignment manufactures change that does not exist. The logged residual doubles as visible proof the check was performed.

4. Stage 1 — Agentic controller

The statement lists six controller duties. Each is implemented as a separately named and separately logged step, so the observable trace maps onto the required list without interpretation.

4.1 Interpret the query and classify the task

Version	Mechanism	Use	Target
v1	keyword and pattern rules over the six intents	internal round; already implemented in the demo server	adequate on the five representative queries
v2	fine-tuned MiniLM or DistilBERT, 6-way, CPU inference	finale; trained on templated seeds expanded by paraphrase	intent accuracy above 97 per cent, with the confusion matrix published

A trained classifier also answers the “no generic LLM” clause more cleanly than prompt engineering does: it is a component that was trained, and its accuracy is reportable. Slots are extracted alongside the intent: **target_class**, **temporal_refs**, **polarity** (increased, decreased, unchanged), **modality_hint**, and **output_kind** (text, box, mask, categorical, number).

4.2 Compatibility gate

The statement names five properties to verify — number, modality, format, metadata, compatibility — so the gate is literally five fields with a logged verdict each, passes included.

Intent	Preconditions	On failure
vqa, describe, ground	at least one tile; readable; profile resolved	reject with a typed error
ground	target class within the supported vocabulary	abstain, and suggest the nearest supported class
change, change-qa	exactly two tiles; distinct timestamps; footprint IoU above 0.9; residual alignment shift under 2 px	abstain, naming the specific unmet condition
fuse	one optical and one SAR tile; co-registered; footprints agree	degrade to single-modality, flagged explicitly

> **Abstention is a feature, not an excuse**
>
> Annotations on the ISRO evaluation set are undisclosed, so a confident wrong answer costs score with no way to detect it beforehand. A reasoned abstention reads as engineering maturity and is measured directly by abstention precision in Stage 6.

4.3 Tool registry and selection

Selection is a declarative filter over specs, never a hardcoded branch:

```
ToolSpec:
  name, version, weights_hash
  tasks: [vqa | describe | ground | change | change-qa | fuse]
  modalities: [optical | sar | optical+sar]
  gsd_range: (min, max)        # lets the controller prefer a model trained near
                               # the input resolution -- and makes the Cartosat
                               # gap visible in the trace instead of silent
  permitted_params: {conf_thresh: [0.1, 0.9], topk: [1, 50], dead_band: [0.0, 0.1]}
  device, load_cost_mb
```

4.4 Parameter binding

The statement requires that the system “configure only permitted task parameters,” so this is enforced rather than documented: the controller may write only keys present in **permitted_params**, every value is range-checked against the declared bounds, and anything else raises. That is a security property and a compliance property in one mechanism.

5. Stage 2 — Specialist models

One uniform signature, enforced by an abstract base class, so a stub and a trained checkpoint are interchangeable and the controller never changes:

```
run(tiles: list[Tile], slots: dict, params: dict) -> TaskResult

TaskResult:
  text            # natural-language answer
  visual          # geo-referenced boxes, masks, or overlays
  confidence      # calibrated in Stage 3, not raw softmax
  raw             # the numbers behind the text, so prose stays auditable
  model_id, model_version, weights_hash
```

5.1 Shared adapted backbone

A RemoteCLIP or SigLIP-class vision encoder, QLoRA-adapted on BigEarthNet's multisensor image-text pairs, with per-task adapters swapped at inference. Four adapters at roughly 50 MB each beat four full checkpoints on a demo machine, and one adapted backbone satisfies the mandatory adaptation clause outright.

5.2 Closing the resolution and sensor gap

Folded into backbone training rather than applied afterwards. This is the single highest-value engineering decision in the document.

- **Multi-scale augmentation.** Random resampling across a 0.3-10 m equivalent range, so the encoder never sees one fixed scale. BigEarthNet upsampled aggressively is a poor proxy for genuine sub-metre detail, so supplement with real high-resolution open data: DOTA, DIOR, LoveDA, Potsdam and Vaihingen. The statement permits “any open source training data,” so this is in scope.
- **GSD conditioning.** Feed the tile's ground sample distance in as a scalar embedding added to the visual tokens. Cheap, and it lets one model behave differently at 0.65 m than at 10 m instead of averaging the two.
- **SAR generalisation.** Train on Sentinel-1 VV/VH with heavy speckle and incidence-angle augmentation. RISAT is C-band like Sentinel-1, which helps, but polarisation availability may differ, so the SAR branch must tolerate a missing second polarisation rather than requiring both.
- **Adversarial validation split.** Hold out a deliberately high-resolution, non-Sentinel split. Validating only on BigEarthNet hides this failure until evaluation day, when nothing can be done about it.

5.3 Task heads

Task	Approach	Training data	Output
vqa mandatory	backbone + projector + small LM, adapter-tuned; handles closed-form types (presence, count, comparison, area) and open text	RSVQA LR/HR train splits; test splits kept clean for reporting	text, categorical
describe	caption head on the shared backbone	BigEarthNet text, VRSBench captions, RSICD	text
ground	open-vocabulary detector for countable classes; CLIPSeg or SAM for areal classes such as water and vegetation, where a box is the wrong primitive	DIOR-RSVG, VRSBench referring expressions	**geo-referenced** boxes and masks, converted through the tile transform
change mandatory	ChangeFormer or BiT over the aligned pair	LEVIR-CD, OSCD, and SECOND — SECOND matters because CDVQA is built on it, so its semantic-change label space is what gets scored	change mask + description

Task	Approach	Training data	Output
change-qa	**No separate model.** Derived from the change mask: class-wise area at t1 and t2, signed delta, then verbalised with a dead-band for “unchanged.” Every CDVQA question type is computable this way, which is more robust than generative answering.	evaluated on CDVQA; no training beyond the change model	categorical + the areas behind it
fuse mandatory	dual-encoder optical + SAR, mid-level feature fusion, UNet decoder	BigEarthNet co-registered S1/S2 pairs	built-up, water, vegetation, bare masks

Captioning and grounding: the statement requires only one of the two, but VRSBench scores captioning, grounding *and* VQA. Both are one shared backbone plus one light head, so building both turns a partial benchmark result into a complete one.

> **State fusion's value as cloud robustness, not as a general claim**
>
> The demonstrable benefit is specific: when the Stage 0 quality mask flags optical occlusion, the fusion branch is what keeps an answer available at all. Show that case directly rather than asserting that two modalities are generically better than one — the ablation in Stage 6 measures exactly this.

6. Stage 3 — Integration, confidence, abstention

6.1 Per-task calibration

Temperature scaling fitted on held-out splits, so that 0.8 means the same thing coming from VQA as from change detection. Raw softmax scores from four different architectures are not comparable, and a confidence display built on uncalibrated scores is decoration rather than information. Expected calibration error is reported per task.

6.2 Cross-model consistency check

When grounding finds no water polygon but VQA answers “yes, water is present,” raise a conflict flag and lower the merged confidence. It is cheap to implement and it is the one behaviour an orchestrated system has that a single monolithic model cannot — which is precisely the novelty the statement claims for the agentic framing. Treat it as a headline capability, not a detail.

6.3 Threshold, abstain, merge

- Below the calibrated threshold, return an explicit insufficient-confidence result carrying the supporting evidence, rather than a guess.
- Every answer carries text and spatial evidence together wherever the task admits both, plus the numbers in **raw** that the text was derived from.
- Merged confidence accounts for gate degradations: a fusion query answered from a single modality reports lower confidence and says why.

7. Stage 4 — Trace and evidence

The statement names the required fields: selected task, model and tool names, key parameters, outputs. So the trace is machine-readable and is the single source of truth from which the live view, the JSON view and the PDF report all render. One source means the report can never disagree with the screen.

```
TraceStep:
  seq, stage                  # 0..6, mirrors the controller duties in the statement
  intent
  tool_name, tool_version, weights_hash
  params                      # only keys from ToolSpec.permitted_params
  input_tile_ids, input_hash  # input_hash + weights_hash == reproducible run
  output_kind, output_summary
  confidence                  # calibrated
  latency_ms
  gate_verdict                # per-field pass/fail, including the passes
```

- **Reproducibility.** Input hash plus weights hash is what an ISRO or SAC reviewer will ask for. Store both on every step.
- **No internal reasoning text.** Explicitly not evaluated, so it is not stored. This keeps the record compact and auditable.
- **Evidence rendering.** Geo-anchored boxes and masks, change overlay, confidence, and the class areas, all read from the same record.

8. Stage 5 — Application and serving

Endpoint	Purpose	Notes
POST /upload	GeoTIFF or benchmark image in; tiles plus manifest out	runs the full Stage 0 sequence and the format whitelist
POST /query	natural-language query plus tile references; answer out	the demo server's existing response contract already matches
GET /report/{run_id}	downloadable PDF report	rendered from the trace record
GET /health	readiness	lists loaded models with versions

- **Model residency.** LRU registry with lazy loading. Four specialists resident simultaneously will not fit a demo machine.
- **Offline mode.** int8 or ONNX quantisation, a pre-seeded tile cache, and vendored JavaScript. Third-party CDN calls for the 3D globe and the imagery basemap are the current offline failure point.

> **Security posture of /upload**
>
> Accepting arbitrary GeoTIFF over HTTP needs a file-size cap, a band-count cap, a pixel-dimension cap, and a decompression-bomb guard — a crafted GeoTIFF can expand enormously on read.
>
> There is no authentication on the endpoint. That is acceptable for a local demonstration, but it should be declared as a known limitation rather than discovered during judging.

9. Stage 6 — Evaluation harness

Because metrics are normalised and then combined, a zero anywhere is expensive. The harness is built before the models so the weakest task is always known, and “it works in the demo” is never the strongest available claim.

Axis	Metric	Dataset / split
Single-image VQA	accuracy broken out by question type	RSVQA LR and HR, prescribed test splits
Captioning	CIDEr, BLEU-4, ROUGE-L	VRSBench
Grounding	accuracy at IoU 0.5	VRSBench referring expressions, DIOR-RSVG
Change detection	F1 and IoU on the change mask	LEVIR-CD, SECOND
Change VQA	accuracy by question type (change-or-not, ratio, dominant class, increase/decrease)	CDVQA

Axis	Metric	Dataset / split
Optical-SAR fusion	per-class IoU for built-up and water	BigEarthNet held-out pairs
Router	intent accuracy, confusion matrix, paraphrase-robustness delta v1 to v2	generated query set with held-out paraphrases
Calibration	expected calibration error, reliability diagram, per task	held-out splits
Abstention	precision of abstention — of the queries declined, what share would have been answered wrongly. This is the number that makes declining a capability rather than an excuse.	all benchmark splits
Generalisation proxy	full task suite on a high-resolution, non-Sentinel holdout. The only pre-evaluation signal available for the Cartosat gap.	DOTA, DIOR, Potsdam holdout

9.1 Ablations tied to specific claims

Claim being defended	Ablation
Optical-SAR fusion earns its complexity	fusion against optical-only, restricted to cloud-occluded tiles
GSD conditioning closes part of the resolution gap	conditioning on against off, evaluated across the 0.15-10 m range
The trained router beats lexical rules	v2 against v1 on held-out paraphrases of the five representative queries
Derived change-QA beats generative change-QA	mask-derived areas against a generative head, on CDVQA

10. Risks, ranked

#	Risk	Impact	Mitigation
1	**Resolution and sensor gap.** BigEarthNet at 10 m against Cartosat-2S sub-metre plus RISAT.	Severe, and invisible until final scoring	Multi-scale augmentation, GSD conditioning, high-resolution open data, and a non-Sentinel holdout built before the models
2	**Cartosat-2S band structure.** PAN is one band, MX is four; neither is RGB nor Sentinel-2's twelve.	Hard failure on evaluation day	Semantic band map in the sensor profile; absent slots stay None rather than being zero-filled
3	**Bi-temporal misalignment** producing phantom change at sub-metre scale.	Silently wrong change answers	Phase-correlation alignment with the residual shift logged and gated
4	**Compute budget** for four specialists on demo hardware.	Demo instability	Shared backbone with swappable adapters; LRU model registry; quantisation
5	**Dataset link.** BigEarthNet.txt is cited as arxiv.org/abs/2603.29630 — **verify it resolves and that the text annotations are downloadable** before committing the adaptation plan.	Blocks the mandatory adaptation clause	Fallback: BigEarthNet-MM with generated captions. The statement also permits any open-source training data
6	Router confusion between **change** and **change-qa**.	Wrong output kind, scored as wrong	Training data weighted toward that boundary; confusion matrix reported

11. Build order

One reordering matters. Stage 0 gates everything: every specialist needs the canonical tile contract before it can train, and the sensor abstraction is what protects the team on evaluation day. Real-tile work is often scheduled last; here it goes first.

Phase	Work	Exit gate
P0	Sensor profile registry, canonical tile contract, pair checks and alignment, tile cache. Evaluation harness scaffolded in parallel.	A Sentinel pair and a synthetic sub-metre pair both ingest to identical Tile objects

Phase	Work	Exit gate
P1	Controller: router v2, five-field compatibility gate, tool registry, parameter binding, trace schema.	All six intents route and gate correctly against stubs; trace validates
P2	Backbone adaptation: QLoRA on BigEarthNet with multi-scale augmentation and GSD conditioning.	Loss curve plus adapter; holdout score on the non-Sentinel split recorded
P3	Task heads: VQA, caption, grounding, change, derived change-QA, fusion.	Each head returns a valid TaskResult; benchmark numbers on prescribed splits
P4	Calibration, cross-model conflict detection, abstention thresholds.	ECE reported per task; abstention precision measured
P5	FastAPI backend, LRU registry, GUI, overlays, report generation, offline packaging.	End-to-end run offline, with a downloadable report matching the on-screen trace
P6	Ablations, final scorecard, submission material.	Every claim in section 9.1 has a number attached

11.1 For the 30 September idea submission

What has to be *written* rather than built, in priority order:

- The agentic architecture diagram and the six-intent registry table (sections 2.1, 2.2, 4.3).
- The trace schema, presented as the answer to the statement's auditability clause (section 7).
- The domain-adaptation story **with the resolution-gap mitigation made explicit** (sections 1.1, 5.2). Most submissions will not notice the gap between the mandated training corpus and the Cartosat evaluation set; a SAC reviewer will recognise it immediately, and addressing it unprompted is the strongest available signal of domain competence.
- The evaluation plan against all four named benchmarks, including abstention precision (section 9).
- Requirements traceability, showing every mandatory clause mapped to a component (section 1.4).

Problem statement text retrieved from sih.gov.in/sih2026PS on 7 September 2026. Requirements in sections 1.4 and 2.2 are quoted or closely paraphrased from that source; everything else is proposed engineering and open to revision. The BigEarthNet.txt dataset link given in the statement is unverified — see risk 5.